

# Agent-Based Modeling of Vaccination Methods to Combat White-Nose-Syndrome

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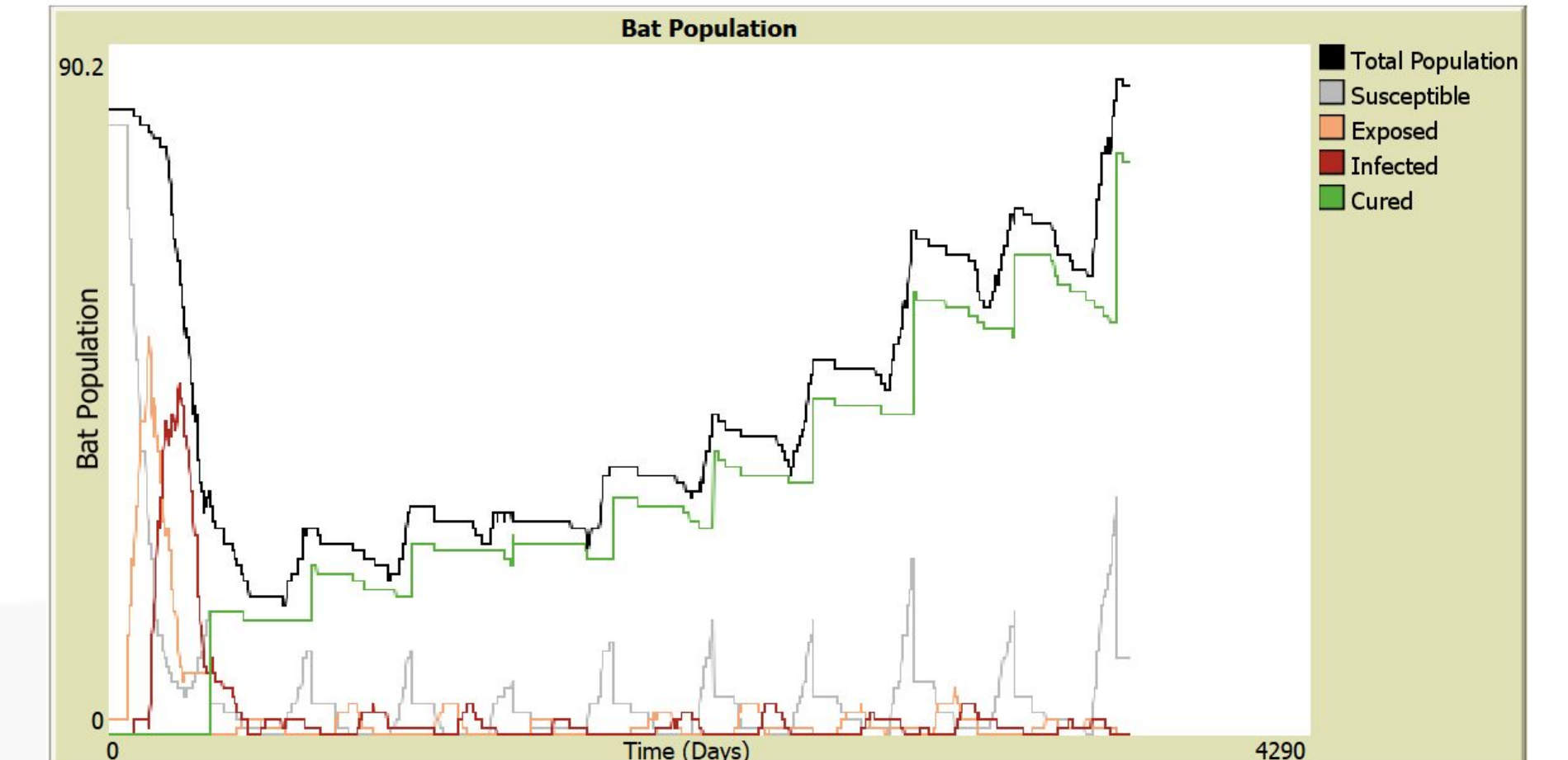
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## Introduction

- White Nose Syndrome (WNS), caused by *Pseudogymnoascus destructans*, severely impacts North American bat populations by spreading in cold, humid caves where bats hibernate, leading to mass mortality.
- This study uses NetLogo to simulate the effects of different vaccination rates, timings, and colony behaviors on disease outcomes, aiming to identify effective strategies for real-world bat conservation.

## Methods

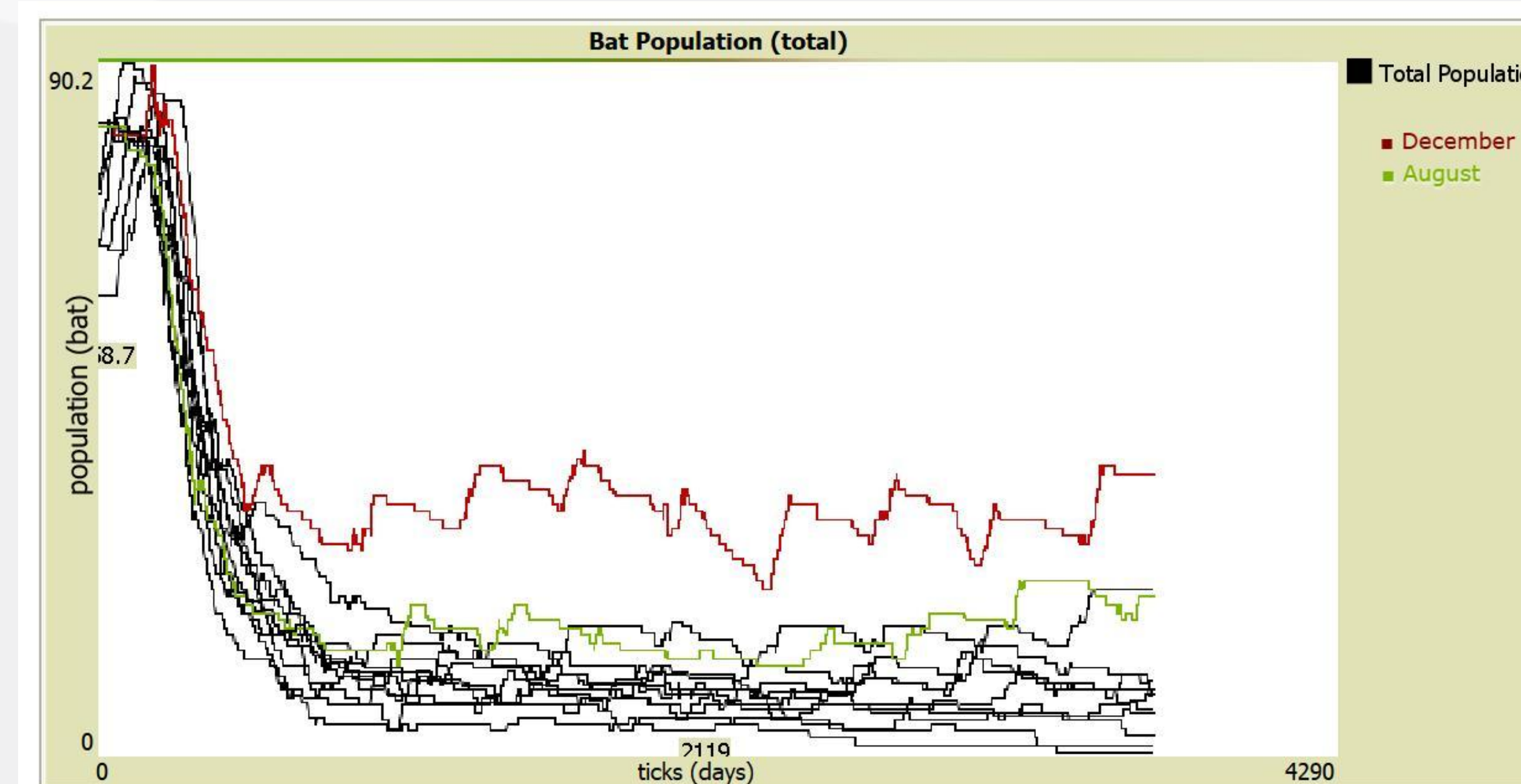
- Developed an agent-based model in NetLogo to simulate bat colony dynamics and the spread of WNS.
  - Represented individual bats as agents with four classes: Susceptible, Exposed, Infected, Cured.
  - Designed scenarios with different timing of vaccination with possible change of vaccination rate.
- Collected output data on infection prevalence, mortality rates, and overall colony survival across simulations.



## Results

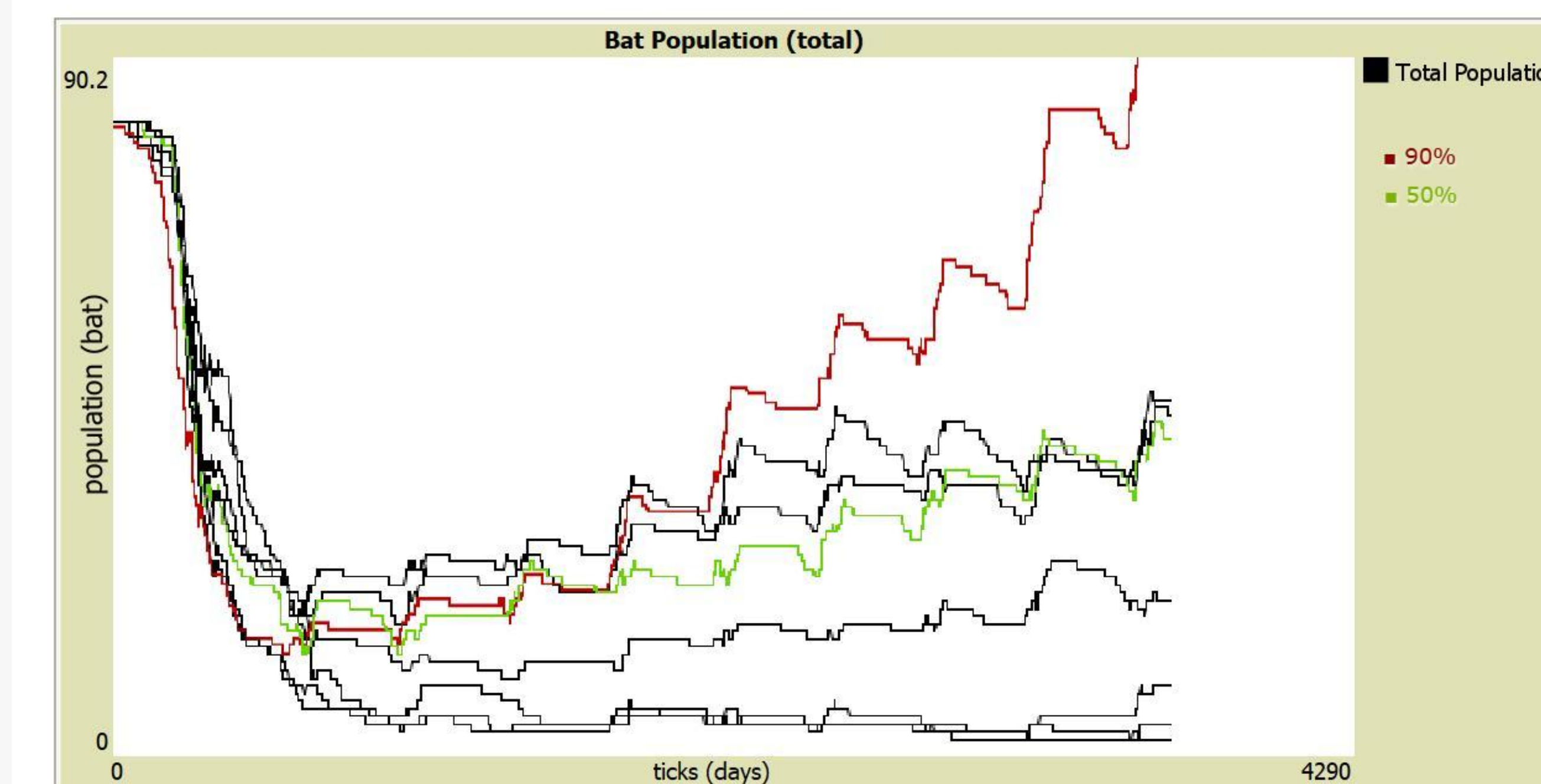
We collected population data with **40%** vaccination rate for 12 different months. Our results showed that December (red,) appeared to be the most efficient month, but it is during the hibernation period for the bats, which means that their immune systems are suppressed.

Next most successful month was August (green,) which is when all the pups are born, and the roosting stage begins.



Our observations showed that when vaccination rates were below **30%**, the bat population collapsed too quickly.

However, once vaccination rates surpassed **40%**, the population began to stabilize. While 40% and 50% vaccination rates were fairly similar in their effects, achieving over **60%** was unrealistic due to logistical challenges and high costs, especially during the hibernation period.



## Future Directions

- Spray vaccine: there exists a method that would offer an alternative to Intramuscular injections and could be far more efficient as it relies on an aerosol that would stick to the bats fur.
- We are yet to incorporate a the environment-to-bat exposure into our model, we plan to add a cave system in net logo which would make it more appropriate to add this variable to the simulation.

## Acknowledgements

- We would like to express our sincere gratitude to the Haynes Program for providing us with the opportunity.
- Additionally, we extend our thanks to the Department of Mathematics and Statistics at James Madison University for their support.
- We also wish to acknowledge Dr. Alex Capaldi and Dr. Laura Tipton for their time and patience

